UNIVERSITY OF MAURITIUS

SIS 3204Y(5) - NATURAL LANGUAGE PROCESSING

Week 5 - 10 Sample Exam Questions

RNN, LSTM and GRU | Sample Answers and Marking Rubrics

These questions are designed as a revision bank based on the Week 5 lecture. They progress from conceptual understanding to application, comparison and analysis.

Question 1 - 6 marks

Explain how a Recurrent Neural Network (RNN) processes the sentence "The movie was excellent". In your answer, explain the role of the hidden state.

Sample Answer

An RNN processes the sentence one word at a time. Each word is first represented numerically, for example using a word embedding. At each time step, the RNN combines the current word representation with the hidden state from the previous time step. The hidden state acts as a memory of the information processed so far. Therefore, after processing "The movie was", the hidden state contains contextual information from these words. When "excellent" is processed, this previous context is combined with the representation of "excellent" to produce a new hidden state. The final hidden state can then be used for an NLP task such as sentiment classification.

Marking Rubric

Expected element	Marks
RNN processes words sequentially	1
Words are represented numerically/using embeddings	1
Current input is combined with previous hidden state	1
Hidden state carries information from previous words	2
Final hidden state can be used for prediction	1
Total	**6**

Question 2 - 6 marks

Explain Backpropagation Through Time (BPTT) and its role in training an RNN.

Sample Answer

During the forward pass, an RNN processes the sequence one time step at a time and produces a prediction. A loss function measures the difference between the predicted and expected output. For training, the RNN is conceptually unrolled across the different time steps. BPTT propagates the error backwards through these time steps. Gradients are calculated to determine how the shared weights contributed to the error. The weights are then updated, for example using gradient descent, so that the RNN can reduce its prediction error.

Marking Rubric

Expected element	Marks
Mentions forward processing/prediction	1
Loss measures prediction error	1
RNN is unrolled through time	1
Error is propagated backwards across time steps	1
Gradients are calculated	1
Weights are updated to reduce loss	1
Total	**6**

Question 3 - 8 marks

What is the vanishing gradient problem? Explain why it makes it difficult for a standard RNN to learn long-term dependencies in text.

Sample Answer

When an RNN is trained using BPTT, gradients are propagated backwards through many time steps. During this process, the gradients may repeatedly become smaller. After many time steps, they can become extremely small. As a result, information from words appearing much earlier in a sequence has very little influence on the weight updates. The RNN therefore struggles to learn relationships between words that are far apart. For example, in “The cat that chased the mouse was hungry”, the model needs to retain information about “cat” when it later processes “was”. If the influence of the earlier word is lost, the RNN may fail to learn this long-term relationship.

Marking Rubric

Expected element	Marks
Relates problem to training/BPTT	1
Explains that gradients become progressively smaller	2
Explains reduced learning from earlier time steps	2
Connects problem to long-term dependencies	1
Appropriate NLP example	1
Correctly explains the example	1
Total	**8**

Question 4 - 8 marks

Explain the role of the forget gate, input gate and output gate in an LSTM. How do these gates help an LSTM process long sequences?

Sample Answer

An LSTM uses gates to control the flow of information through its memory. The forget gate determines which information from the previous cell state should be retained or discarded. The input gate determines which new information should be added to the cell state. The output gate determines which information from the updated cell state should contribute to the current hidden state. These gates allow the LSTM to selectively remember useful information and remove information that is no longer relevant. Together with the cell state, this enables an LSTM to preserve important information over longer sequences and reduces the long-term dependency problem experienced by a standard RNN.

Marking Rubric

Expected element	Marks
Correct explanation of forget gate	2
Correct explanation of input gate	2
Correct explanation of output gate	2
Explains selective information/memory control	1
Connects gates to long-term dependencies	1
Total	**8**

Question 5 - 6 marks

Distinguish between the hidden state and cell state of an LSTM.

Sample Answer

The cell state acts as the longer-term memory of the LSTM. It carries selected information across time steps and is controlled mainly through the forget and input mechanisms. The hidden state represents the current contextual output of the LSTM at a particular time step. It is influenced by the cell state and the output gate and can be used to make predictions as well as being passed to the next time step. Therefore, the cell state mainly provides a long-term memory path, while the hidden state represents the current contextual information.

Marking Rubric

Expected element	Marks
Identifies cell state as longer-term memory	2
Explains that cell state carries information across time	1
Identifies hidden state as current contextual representation	1
Connects hidden state to output/prediction	1
Clear distinction between the two	1
Total	**6**

Question 6 - 6 marks

A GRU has two gates: the reset gate and the update gate. Explain the function of each gate.

Sample Answer

The reset gate controls how much information from the previous hidden state should be considered when the GRU creates its new candidate hidden state. A low reset value causes more previous information to be ignored, whereas a higher value allows more past information to contribute. The update gate controls the balance between retaining the previous hidden state and using the new candidate information. It therefore determines how much the GRU should preserve from the past and how much it should update using the current input. Together, the two gates allow the GRU to manage its memory without requiring a separate cell state.

Marking Rubric

Expected element	Marks
Correct role of reset gate	2
Explains effect on previous information	1
Correct role of update gate	2
Explains combined memory-control function	1
Total	**6**

Question 7 - 10 marks

Compare RNN, LSTM and GRU with respect to their memory mechanism, gates, ability to capture long-term dependencies and computational complexity.

Sample Answer

A standard RNN has a single hidden state and no gates. It has the simplest architecture and relatively few parameters, but it is vulnerable to vanishing gradients and therefore struggles with long-term dependencies. An LSTM has both a hidden state and a separate cell state. It uses three gates: forget, input and output. These mechanisms give it greater control over information and allow it to capture long-term dependencies more effectively. However, its architecture is more complex and requires more parameters and computation. A GRU has a single hidden state and uses two gates: reset and update. It therefore has a simpler architecture than an LSTM and generally requires fewer parameters. It can still capture long-term dependencies effectively and often trains faster than an LSTM. The choice depends on the task, sequence characteristics, available computational resources and empirical performance.

Marking Rubric

Expected element	Marks
Correct RNN architecture/memory	1
RNN long-term dependency limitation	1
Correct LSTM states	1
Correct LSTM gates	1
LSTM long-term memory advantage	1
LSTM complexity	1
Correct GRU state and gates	2
GRU efficiency/simpler architecture	1
Meaningful overall comparison	1
Total	**10**

Question 8 - 10 marks

You are developing a system to classify long customer reviews as positive, neutral or negative. Some reviews contain important information near the beginning that affects the interpretation of words much later in the text. Discuss whether an RNN, LSTM or GRU would be appropriate for this task.

Sample Answer

A standard RNN could perform sentiment classification, but it may not be the most suitable architecture for long reviews because important information may need to be retained across many time steps. Vanishing gradients can make it difficult for an RNN to learn these long-term dependencies. An LSTM is designed to preserve information for longer periods through its cell state and forget, input and output gates. It is therefore suitable when long-term relationships in the reviews are important, although it has more parameters and greater computational cost. A GRU also provides gated memory using reset and update gates. It is simpler than an LSTM and normally has fewer parameters while still being able to model long-term dependencies. Both LSTM and GRU would therefore be reasonable architectures to evaluate, with the final selection based on computational resources and validation performance.

Marking Rubric

Expected element	Marks
Identifies long-term dependency requirement	1
Explains RNN limitation	2
Explains why LSTM is suitable	2
Discusses LSTM complexity	1
Explains why GRU is suitable	2
Discusses GRU efficiency	1
Justified model-selection approach	1
Total	**10**

Question 9 - 8 marks

Describe four NLP applications in which recurrent neural network architectures can be used. For each application, briefly explain what the model learns or predicts.

Sample Answer

Examples include: (1) Language modelling - the model learns patterns in sequences and predicts the next word. (2) Text classification - the model represents a text sequence and predicts a category such as sentiment. (3) Named Entity Recognition - the model assigns labels such as PERSON, LOCATION or ORGANISATION to tokens. (4) Machine translation - the model processes a sequence in one language to generate a corresponding sequence in another language. Other valid lecture examples include part-of-speech tagging, text summarisation, dialogue systems/chatbots and question answering.

Marking Rubric

Expected element	Marks
Valid NLP application 1 + correct explanation	2
Valid NLP application 2 + correct explanation	2
Valid NLP application 3 + correct explanation	2
Valid NLP application 4 + correct explanation	2
Total	**8**

Question 10 - 10 marks

RNN, LSTM and GRU have been widely used for NLP. Explain the limitations of recurrent architectures that motivated the move towards attention and Transformers.

Sample Answer

RNNs, LSTMs and GRUs process text sequentially. Each time step depends on information from previous time steps, which limits the amount of parallel processing that can be performed. A vanilla RNN also has difficulty learning long-term dependencies because of vanishing gradients. LSTM and GRU improve this through gating mechanisms, but they still process sequences recurrently and can become inefficient for very long texts. Attention allows a model to determine which words or positions in a sequence are most relevant to one another, so distant information can be accessed more directly. Transformers build on self-attention and remove the need for recurrent token-by-token processing. This enables greater parallelisation and makes them more scalable for large datasets and long sequences.

Marking Rubric

Expected element	Marks
Identifies sequential nature of recurrent processing	2
Explains long-term dependency/gradient limitation	2
Recognises that LSTM/GRU improve but do not remove all limitations	1
Correctly explains basic idea of attention	2
Explains removal of recurrence in Transformers	1
Explains parallelisation/scalability advantage	2
Total	**10**